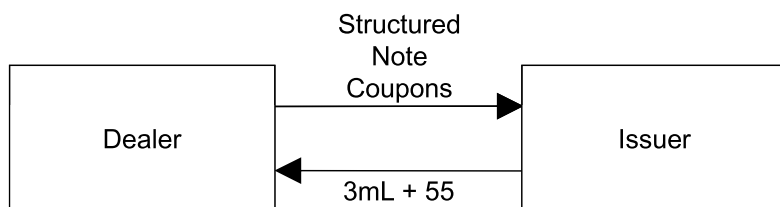


Accrual Start	Accrual End	LIBOR Forward	Accrual Start	Accrual End	LIBOR Forward
20-Sep-03	20-Dec-03	1.15%	20-Sep-08	20-Dec-08	6.02%
20-Dec-03	20-Mar-04	1.27%	20-Dec-08	20-Mar-09	6.06%
20-Mar-04	20-Jun-04	1.56%	20-Mar-09	20-Jun-09	6.13%
20-Jun-04	20-Sep-04	1.99%	20-Jun-09	20-Sep-09	6.17%
20-Sep-04	20-Dec-04	2.49%	20-Sep-09	20-Dec-09	6.19%
20-Dec-04	20-Mar-05	2.97%	20-Dec-09	20-Mar-10	6.23%
20-Mar-05	20-Jun-05	3.38%	20-Mar-10	20-Jun-10	6.28%
20-Jun-05	20-Sep-05	3.75%	20-Jun-10	20-Sep-10	6.31%
20-Sep-05	20-Dec-05	4.05%	20-Sep-10	20-Dec-10	6.31%
20-Dec-05	20-Mar-06	4.34%	20-Dec-10	20-Mar-11	6.31%
20-Mar-06	20-Jun-06	4.56%	20-Mar-11	20-Jun-11	6.31%
20-Jun-06	20-Sep-06	4.78%	20-Jun-11	20-Sep-11	6.32%
20-Sep-06	20-Dec-06	4.99%	20-Sep-11	20-Dec-11	6.35%
20-Dec-06	20-Mar-07	5.19%	20-Dec-11	20-Mar-12	6.38%
20-Mar-07	20-Jun-07	5.35%	20-Mar-12	20-Jun-12	6.41%
20-Jun-07	20-Sep-07	5.51%	20-Jun-12	20-Sep-12	6.45%
20-Sep-07	20-Dec-07	5.64%	20-Sep-12	20-Dec-12	6.49%
20-Dec-07	20-Mar-08	5.77%	20-Dec-12	20-Mar-13	6.52%
20-Mar-08	20-Jun-08	5.87%	20-Mar-13	20-Jun-13	6.55%
20-Jun-08	20-Sep-08	5.96%	20-Jun-13	20-Sep-13	6.58%

Table 7.17: 3-Month LIBOR Forwards Based on the Swap Curve in Figure 7.17



Notional:	\$100mm
Maturity:	10 years
Dealer Pays:	Structured Note Coupons
Structured Note Coupons:	Year 1: 10%
	Years 2–10:
	10% – 2 x 3-month LIBOR, floored at 0%
Issuer Pays:	3-month LIBOR + 55 bps
Knockout Feature:	If the sum of accrued coupons paid by the Dealer ever exceeds the Lifetime Target, then the swap “knocks out”
Lifetime Target:	13.05%

Figure 7.18: The Swap Transaction Implicit in the Target Redemption Note

transaction, this implies that the (off-market) value of the high coupon plain vanilla swap must be exactly offset by the value of the embedded receiver swaption. In a similar manner, Figure 7.19 shows how we can decompose the transaction shown in Figure 7.18 into a (relatively) plain vanilla swap in which the dealer pays the structured note coupons versus 3-month LIBOR + 55 bps for ten years, and an option. If this option finishes in-the-money, it will cause the transaction to end early.

In Figure 7.19, the top transaction is the knockout swap shown in Figure 7.18. The middle line in Figure 7.19 represents the relatively plain vanilla swap in which the dealer pays the structured note coupons versus 3-month LIBOR + 55 bps for ten years, and the bottom line represents the knockout feature that potentially causes the trade to end prior to year 10. Since the

that the dealer is attributing to this knockout feature. In other words, if we were assigned the task of putting this structured note together, we would want to compute the price of the knockout feature itself (presumably using some sophisticated model). Our goal here is to take the pricing of the note as a given and see if we can back out the price of the knockout feature as a function of other prices that are embedded in the note.

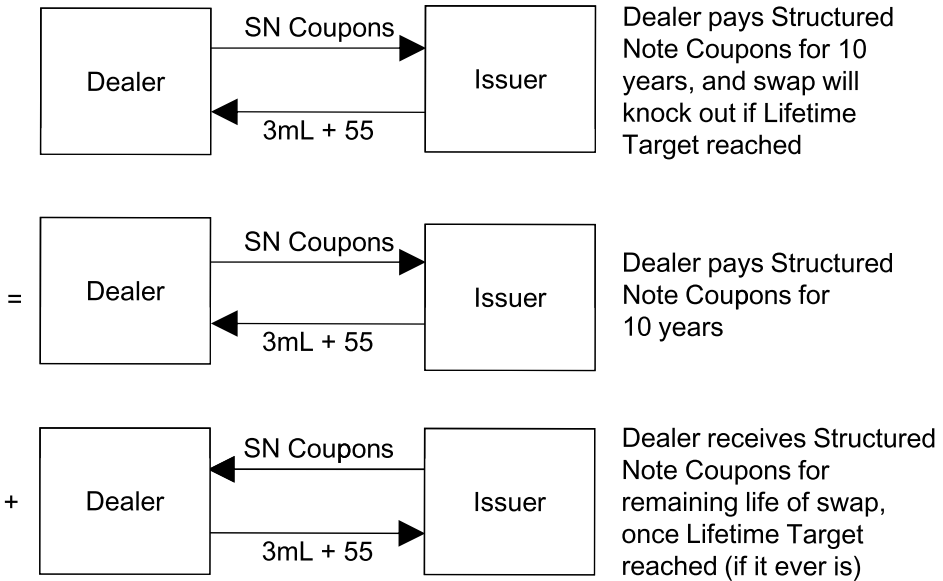


Figure 7.19: Decomposing the Knockout Swap

top transaction is an $NPV = 0$ transaction, this implies that the value of the middle transaction must be exactly offset by the value of the knockout feature in the bottom transaction.³⁵

Finding the value of the middle transaction in Figure 7.19 is not all that difficult, however. Based on a proprietary model, the swap curve shown in Figure 7.17 prevailing at the time of pricing, and implied caplet volatilities at the time (which are not shown here), some prices (based on a notional of \$100mm) have been computed and are shown in Table 7.18.³⁶

It is important to note that the present value of *receiving* coupons of the form $10\% - 2 \times 3\text{-month LIBOR}$ (without the minimum coupon floor of 0%) between years 2 and 10 is negative, as shown in Table 7.18. The reason for this is that although today's LIBOR is 1.14% and the forwards initially start out low, eventually the forward LIBORs are greater than 5% (as shown

³⁵We are assuming throughout this problem that this transaction has been done at mid-market, and hence the swap in Figure 7.18 is an $NPV = 0$ transaction.

³⁶You may assume that all prices presented here are scalable, meaning that if you double the notional, you double the price.

3-month LIBOR stream for 10 years:	\$40,685,181
55 bps annuity for 10 years:	\$4,441,706
10% coupon quarterly for 1 year:	\$9,908,136
10% – $2 \times$ 3-month LIBOR coupons quarterly between years 2–10:	–\$7,375,616
1x10 5% cap:	\$7,240,000

Table 7.18: Some Prices Based on the Swap Curve in Figure 7.17 (assumed notional is \$100mm)

in Table 7.17), at which point the quantity $10\% - 2 \times$ 3-month LIBOR becomes negative.

e) Given the data in Table 7.18, what is the value of the middle transaction in Figure 7.19?

f) Given your answer to part e), what is the value of the exotic path dependent knockout feature depicted by the bottom transaction in Figure 7.19? Who is benefiting from this knockout feature, the swap dealer or the issuer? Do you suspect that this knockout feature is in-the-money or out-of-the-money?

Back to the Bond

Let's take a look at the bond again and consider the particular case in which the forwards are realized. That is, let us assume that over the course of time the LIBOR settings that occur are those that are equal to the forwards. This is, of course, just one of an infinite number of paths that may actually occur over the life of this trade. But it is a path that we often focus on, inasmuch as the forwards (loosely) represent the market's expectation of where LIBOR will be in the future.

g) Make a chart that shows the projected coupons as predicted by the forwards through time. That is, assuming that the forwards are realized, provide a chart that shows the projected coupons paid by the bond as a function of time.

h) Refer back to the terms of the bond as shown in Table 7.15. Suppose for the moment that the "Early Redemption Provision" as specified in Table 7.15 is not applicable. That is, suppose that the terms of the bond are as

given in Table 7.15, except there is no Early Redemption Provision. What are the projected coupons paid by the bond as a function of time under this assumption? Take your chart that you just made in part g) and show on the same chart the projected coupons both with and without the Early Redemption Provision.

i) Does the existence of the Early Redemption Provision in the terms of the note increase or decrease the projected coupons according to the forwards? How do you reconcile this finding with your intuition about what the Early Redemption Provision does for the bondholder?

j) If you had purchased this note back in August 2003, how would you have done on it, given the realizations of LIBOR that actually occurred over time?

Chapter 8

Relative Value and Macro Trades

It's impractical to completely categorize all the types of speculative trades that people put on in the swaps market—there are just too many of them. But what we will try to do in this chapter is give a sense of some of the major categories of trades that people often look at. Please bear in mind from the outset that this will be far from complete.

Broadly speaking, we can classify speculative trades as being either *relative value* or *macro* in nature. By relative value, we refer generically to trades that involve “buying” one or more components (which are thought to be relatively cheap) and simultaneously “selling” other components (which are thought to be relatively rich) in hopes that over time the long position will enrich relative to the short position, and hence the trade will make money.¹ Macro trades refer to trades that involve an outright long or short position in one component, thereby expressing a view that the price of something will go up or down over time.

Many investors believe that over the years “good” relative value trades have been harder and harder to find.² That is, trades in which there is a blatant mispricing between the components are a lot less evident. The reason for this is that over time the markets have become more efficient as more and more

¹The words “buying” and “selling” are in quotes because when these components are, say, swaps, there is no bona fide buying or selling involved but rather the paying of or receiving of fixed.

²This was not the case after Lehman fell during the financial crisis. Many investors believed there were very attractive relative value trading opportunities from the fall of 2008 into 2010 due to dislocations across markets.

people have been looking for relative value opportunities to trade, dissipating pricing discrepancies otherwise found in the market. In fact, these days many trades that people refer to as (pure) relative value trades are hardly that. They may contain some other sources of risk, such as having some macro bent to them or having *basis risk*.³

In this chapter we will take a look at a variety of trades that people put on in order to take a view on, for example, movements in swap spreads, the shape of the swap curve, or asset swap levels. We will also introduce the notion that sometimes people use options to exploit opportunities found in the outright market. That is, by buying certain (cheap) options and selling other (rich) options, one might be able to put on a trade that is (perhaps) more attractive than what could otherwise be achieved by trading in the underlyings of those options. An important point to keep in mind, however, is that when one uses options to enhance the terms of a trade, this can sometimes lead to the original trade view becoming obfuscated. In addition, it introduces new risks (e.g., exposure to movements in implied volatility) that must be considered.

8.1 Carry and Roll-Down Analysis

Oftentimes when investors put on speculative trades, they expect to hold the position for some period of time until their objective in the market is achieved. They may firmly believe that their view will come to fruition but are uncertain as to the exact timing. Investors often opt to put on a trade immediately, rather than risk missing out on the opportunity if they wait on the sidelines. During the time period that a trade is held, it may produce cash flows (or accrue interest) that will contribute to the overall P&L of the trade. In addition, even if the market is unchanged (e.g., if swap rates do not change), the value of the (residual) position may change purely as a function of the passage of time. A common metric employed in the fixed income markets to capture both of these phenomena is referred to as *carry and roll-down analysis*.

We introduced the notion of carry in Chapter 3. For a given horizon, *carry* is defined as the net interest that is earned from holding a trade. Components of carry may include, for example, cash flows from a swap, coupon interest

³*Basis risk* refers to the residual risk left in a trade that consists of a long position in one asset that is hedged by a short position in another asset. For example, if we buy mortgages and hedge the duration of this position by paying fixed in swaps, we will not have eliminated all of the risk in our long position (e.g., we will still be exposed to movements in volatility, prepayments, etc.). This residual risk is referred to as basis risk. In fact, putting on this type of trade, either buying or selling mortgages and hedging the duration by paying or receiving in swaps, is referred to as *trading the mortgage basis*.

on a Treasury, and the cost of financing a Treasury position. Carry includes not only coupon cash flow that is paid out but also any accrued interest earned since the trade's inception.

Let's turn our attention now to a discussion of carry and roll-down as it pertains to swaps. For any given swap and any stated investment horizon, we can partition the cash flows of the swap into the carry earned over the investment horizon and the residual cash flows remaining in the swap. Let (today's) *forward price* denote the present value of the residual cash flows remaining in the swap as of the horizon. Then it must be that

$$\begin{aligned} &\text{Spot Price of Swap} \\ &= \text{Present Value of Carry} + \text{Forward Price of Residual Swap}, \quad (8.1) \end{aligned}$$

where the *spot price* denotes the present value of the whole swap today. This equation holds true for any swap, not just a par swap whose coupon makes the trade an $\text{NPV} = 0$ transaction. Sometimes the three prices in Equation 8.1 may be expressed in dollars, and at other times they may be expressed equivalently in bps of the notional. The logic behind Equation 8.1 is depicted in Figure 8.1.

Any par swap that we enter into today is by definition an $\text{NPV} = 0$ transaction, which is to say that its spot price is zero. Thus by Equation 8.1 it follows that for an on-market swap that we enter into today, and for any given investment horizon, we have

$$\text{Present Value of Carry} = - \text{Forward Price of Residual Swap}. \quad (8.2)$$

Equation 8.2 is very powerful. It lets us calculate the forward price for any par swap given the present value of the carry, or vice versa. In addition, as a consequence of this equation, the carry and the forward price of the residual swap must be of opposite sign. As we will see shortly, this is a useful thing to remember when thinking about trades.

Consider once again our stylized example in which the swap curve is as in Table 8.1 and assume we enter into a 4-year swap in which we receive fixed. Given a horizon of one year, the carry on the trade is $5.47\% - 5.00\% = 47$ bps, the difference between the fixed rate of 5.47% and spot 1-year LIBOR, which is 5.00%. This cash flow is received at the end of the first year. Recall that the 1-year zero coupon rate for this curve is 5%, and thus the present value of the carry is $47 \text{ bps} / 1.05 \approx 44.76$ bps, or \$447,619 on an assumed notional of \$100mm.

Of course, a 4-year swap with a par coupon of 5.47% is an $\text{NPV} = 0$ transaction, which is to say that its spot price is zero. Then according to Equation 8.2, the forward price of the residual swap should be -44.76 bps,

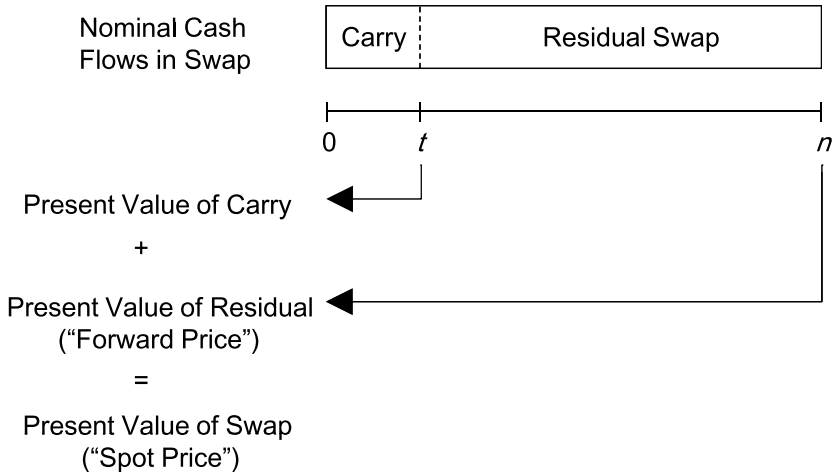


Figure 8.1: An n -Year Swap Partitioned into Carry Over an Investment Horizon t and Residual Cash Flows

or $-\$447,619$ to the fixed rate receiver. The residual swap in this case is a 3-year swap, one year forward (i.e., a 1x4) with a coupon of 5.47%. As Figure 8.2 shows, the value of this swap to the fixed rate receiver is indeed $-\$447,619$. *So I got that goin' for me, which is nice.*⁴

To summarize our findings here with our 4-year swap, the present value of the carry over the horizon was positive, and we found that the present value of the residual swap, or its forward price, was negative. And the value of the carry is exactly offset by the value of the residual swap, since the swap itself is an $NPV = 0$ trade.

⁴A quote from the character Carl Spackler, played by Bill Murray, in the movie *Caddyshack*, the entirety of which merits repeating:

A looper, you know, a caddy, a looper, a jock. So, I tell them I'm a pro jock, and who do you think they give me? The Dalai Lama, himself. Twelfth son of the Lama. The flowing robes, the grace, bald ... striking. So, I'm on the first tee with him. I give him the driver. He hauls off and whacks one—big hitter, the Lama—long, into a ten-thousand foot crevasse, right at the base of this glacier. Do you know what the Lama says? Gunga galunga ... gunga, gunga-lagunga. So we finish the eighteenth and he's gonna stiff me. And I say, "Hey, Lama, hey, how about a little something, you know, for the effort, you know." And he says, "Oh, uh, there won't be any money, but when you die, on your deathbed, you will receive total consciousness." So I got that goin' for me, which is nice.

Excerpt from the movie *Caddyshack*, released on July 25, 1980. Used by permission of Warner Bros.

Maturity	Treasury Yield	Swap Spread	Swap Rate (Ann 30/360)
1	4.80%	20	5.00%
2	4.88%	23	5.11%
3	5.05%	25	5.30%
4	5.19%	28	5.47%
5	5.25%	35	5.60%

Table 8.1: The Swap Curve Today

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	0	0	0	0
2	5.113%	5.226%	5,470,000	4,950,807	(5,225,748)	(4,729,739)
3	5.312%	5.712%	5,470,000	4,683,288	(5,712,202)	(4,890,656)
4	5.494%	6.041%	5,470,000	4,416,473	(6,041,364)	(4,877,792)
				14,050,568		(14,498,187)
		Fixed rate	5.470%	NPV	(447,619)	
		Notional	100,000,000			

Figure 8.2: Value of a \$100mm 1x4 Swap with a Coupon of 5.47%

Let's look at this another way. Suppose we enter into a 4-year swap in which we receive fixed, hold the trade for a year, and unwind it thereafter. How much money should we expect to make with this strategy? We should expect to make zero profits from this strategy. When we enter into the 4-year swap and hold it for a year, we will lock in carry of 47 bps, which has a present value of \$447,619. How much should we expect to make when we unwind the swap at that point in time? In one year, the swap will be a 3-year swap with a coupon of 5.47%. The market's expectation of the 3-year swap rate in one year is the par rate on a 1x4 rate, or the 3-year swap rate, one year forward. We have shown previously in Chapter 3 that for the swap curve in Table 8.1 the 1x4 rate is 5.644%. Receiving on the 1x4 at a rate of 5.47% is therefore receiving a below market coupon, the present value of which is -\$447,619 to us as we saw in Figure 8.2.

For a given investment horizon, the *roll-down* of a swap is defined as the value of the residual swap assuming that the swap curve prevailing at the